Causal Representation Learning and Causal Discovery in Deep Learning: A Comprehensive Survey

Abstract

The integration of causal inference with deep learning has emerged as a pivotal paradigm shift in artificial intelligence, addressing the fundamental limitations of purely correlational models. While deep neural networks excel at capturing complex statistical patterns in high-dimensional data, they often fail to generalize under distribution shifts, lack interpretability, and struggle with counterfactual reasoning. This survey provides a comprehensive review of the state-of-the-art in Causal Representation Learning (CRL) and Causal Discovery within deep learning frameworks. We synthesize findings from 300 recent academic papers to construct a unified taxonomy of methodological approaches, ranging from differentiable structure learning and latent variable identification to causal generative modeling and intervention design. We critically analyze the theoretical foundations of identifiability, examining how assumptions such as Independent Causal Mechanisms (ICM), sparsity, and multi-environment variability enable the recovery of true causal structures from observational and interventional data. Furthermore, we explore the application of these methods across diverse domains including biomedicine, robotics, climate science, and time-series analysis. Finally, we discuss persistent challenges, including the scarcity of real-world causal benchmarks, the computational complexity of exact inference, and the integration of large language models for causal knowledge extraction, outlining a roadmap for future research towards robust and interpretable artificial intelligence.

1. Introduction

The rapid advancement of deep learning has revolutionized fields ranging from computer vision to natural language processing. However, standard deep learning models operate primarily on the first rung of Pearl's Ladder of Causation—association—learning statistical correlations that are often spurious and sensitive to environmental changes [1]. Consequently, these models frequently fail when deployed in out-of-distribution (OOD) settings where the data generating process shifts, a phenomenon known as the "generalization gap" [2]. To overcome these limitations, the research community has increasingly turned to causal inference, seeking to learn representations that capture the underlying invariant mechanisms of the data generating process rather than mere surface-level statistics [3].

Causal Representation Learning (CRL) aims to map low-level, high-dimensional observations (e.g., pixels, text tokens) to high-level causal variables that interact according to a structural causal model (SCM) [4]. Simultaneously, Causal Discovery seeks to infer the graphical structure (Directed Acyclic Graphs or DAGs) governing these variables directly from data [5]. The convergence of these fields with deep learning has given rise to "Neural Causal Models," which leverage the expressive power of neural networks to model non-linear causal mechanisms while adhering to causal constraints [1].

The evolution of this field can be characterized by a transition from discrete, combinatorial search algorithms to continuous, differentiable optimization frameworks. Early methods relied on constraint-based tests or score-based searches that scaled poorly with dimensionality [5]. Recent innovations have transformed causal discovery into a continuous optimization problem by relaxing acyclicity constraints into smooth functions, enabling the use of gradient-based learning [3]. Similarly, CRL has progressed from linear factor analysis to non-linear identifiability proofs leveraging interventions, temporal dynamics, and multi-view data [6].

This survey synthesizes the extensive literature on causal deep learning, organizing the field into coherent methodological families. We begin by establishing the theoretical underpinnings of identifiability and the role of inductive biases. We then provide a detailed examination of methodological advances in causal structure learning, latent causal representation, and causal generative modeling. Special attention is paid to the integration of domain knowledge, the handling of time-series and multimodal data, and the emerging role of Large Language Models (LLMs) in causal reasoning. By aggregating evidence from 300 distinct studies, we aim to provide a rigorous technical overview that highlights both the transformative potential and the current bottlenecks of causal deep learning.

2. Method

The methodological landscape of causal deep learning is vast, encompassing techniques for discovering causal graphs, learning causal representations, and performing causal inference using deep neural architectures. We organize this section into several key thematic areas: Differentiable Causal Structure Learning, Identifiable Causal Representation Learning, Causal Generative Modeling and Counterfactuals, Temporal and Dynamic Causal Discovery, and the Integration of Prior Knowledge and Foundation Models.

2.1 Differentiable Causal Structure Learning

A major breakthrough in causal discovery has been the reformulation of the discrete graph search problem into a continuous optimization task. Traditional algorithms like PC or GES operate in a combinatorial space, making them intractable for high-dimensional data. Deep learning approaches address this by parameterizing the adjacency matrix and enforcing acyclicity via differentiable constraints.

2.1.1 Continuous Optimization and Acyclicity Constraints The seminal shift towards continuous optimization involves relaxing the discrete acyclicity constraint into a smooth function $h(A) = 0$, such as the trace exponential function used in NOTEARS and its variants [5]. This allows the use of gradient descent to learn the graph structure alongside functional mechanisms. Recent works have extended this to non-parametric settings. For instance, RKHS-DAGMA utilizes Reproducing Kernel Hilbert Spaces to approximate non-parametric structural equations, defining edge weights via the L_2 norms of partial derivatives of kernel functions [7]. Similarly, methods like DCD-FG restrict the search space to factor Directed Acyclic Graphs (f-DAGs), linking causal discovery to Boolean matrix factorization to achieve linear-time acyclicity scoring for large-scale biological networks [8].

To improve scalability and stability, newer approaches employ alternative relaxation strategies. DiffIntersort reformulates causal ordering using the Sinkhorn operator to approximate doubly stochastic matrices, enabling gradient-based optimization of causal orders while preserving theoretical guarantees [9]. GFlowCausal transforms the search into a sequential edge generation problem using Generative Flow Networks, employing transitive closure masks to guarantee acyclicity in $O(1)$ time during state transitions [10]. Furthermore, optimization-free methods like SSTS (Score-Schur Topological Sort) decouple density estimation from structure learning, mapping iterative graph marginalization to Schur complement operations on the Score-Jacobian Information Matrix, achieving significant speedups over gradient-based methods [11].

2.1.2 Supervised and Amortized Causal Discovery Another paradigm involves treating causal discovery as a supervised learning problem, where models are trained on synthetic data to predict graph structures from observational samples. DAG-EQ utilizes permutation-equivariant neural networks to map correlation matrices to adjacency matrices, ensuring that the model respects the symmetry of variable indexing [12]. Meta-learning approaches, such as meta-CGNN, encode dataset distributions into task features using Conditional Mean Embeddings, allowing a single network to generalize across diverse causal mechanisms with few samples [13].

Amortized inference has also gained traction for rapid discovery. ACD (Amortized Causal Discovery) employs an encoder-decoder framework where a GNN-based encoder predicts causal graph edges as a distribution, enabling cross-sample generalization for systems sharing common dynamics [14]. CauScale introduces a two-stream encoder with tied axial attention to scale causal discovery to thousands of nodes, compressing observational samples via pooling to retain causal signals while reducing computational footprint [15]. Additionally, LLM-initialized methods like LLM-DCD and GUIDE leverage Large Language Models to provide structural priors or warm-starts for gradient ascent, significantly reducing the search space and improving convergence on real-world datasets [16, 17].

2.1.3 Handling Latent Confounders and Non-Linearities Real-world data often involves unobserved confounders and non-linear relationships. Methods like DeFuSE (Deconfounded Functional Structure Estimation) address non-linear DAGs with correlated Gaussian errors by iteratively identifying nodes via topological depth and normality tests on residuals, separating causal functions from linear confounding under sublinear growth assumptions [18]. For latent variable models, NCFA (Neuro-Causal Factor Analysis) infers an Unconditional Dependence Graph via pairwise independence tests and constructs a Minimum MCM graph to constrain a VAE decoder, providing nonparametric identifiability guarantees [19].

In the presence of latent confounders represented by bidirected edges, Modular-DCM decomposes the causal graph into c-components and trains sub-networks sequentially using adversarial loss, preserving latent confounding dependencies within components [20]. Neural ADMG Learning (N-ADMG) parameterizes magnified SCMs using neural autoregressive flows, proving identifiability for non-linear ADMGs under bow-free assumptions [21]. Furthermore, GRACE-VAE integrates Graph Neural Networks into VAE encoders to process heterogeneous network entities, improving latent causal graph recovery in biological networks by leveraging structured context [22].

2.2 Identifiable Causal Representation Learning

A central challenge in CRL is identifiability: determining whether the true latent causal factors and their mechanisms can be uniquely recovered from observed data. Theoretical results indicate that without additional assumptions or data regimes (e.g., interventions, temporal dynamics), non-linear ICA is generally unidentable [6].

2.2.1 Identifiability via Interventions and Multi-Environment Data Interventions provide the strongest signal for identifiability. CITRIS (Causal Identifiability from Temporal Intervened Sequences) demonstrates that interventions enable the identification of minimal causal variables containing only manipulable information, separating them from invariant components [23]. Extending this, iCITRIS generalizes identifiability to settings with instantaneous causal effects using partially-perfect interventions that remove instantaneous parents [24]. For soft interventions, ILANM (Identifiable Latent Additive Noise Models) proves that identifiability is ensured if causal influence derivatives vary sufficiently across environments [25].

Multi-environment data, where distributions shift due to mechanism changes, also facilitates identification. The Sparse Mechanism Shift hypothesis posits that distribution changes are localized to specific mechanisms, allowing the isolation of causal factors [2]. Methods like iCaRL (invariant Causal Representation Learning) extend iVAE to general non-factorized priors, utilizing score matching to identify latents up to simple transformations across multiple environments [26]. Similarly, CRL-General-Env leverages third-order derivatives of log-densities to distinguish causal directions beyond moral graphs, proving identifiability for nonparametric mixing without known intervention targets [27].

Recent work has relaxed the requirement for known intervention targets. Nonparametric CRL with Unknown Intervention Targets establishes that observational data plus one perfect intervention per node suffices for bivariate identifiability under a genericity condition [28]. Additionally, G-CaRL (Grouped Causal Representation Learning) achieves identifiability in purely observational settings by partitioning variables into non-overlapping groups and using group-wise shuffling as a contrastive pretext task to break inter-group dependencies [29].

2.2.2 Disentanglement via Mechanism Sparsity and Independence The Principle of Independent Causal Mechanisms (ICM) serves as a core inductive bias. ICM-VAE models causal mechanisms as nonlinear diffeomorphic flow-based functions, enforcing independent mechanisms through a structured prior to achieve identifiability up to permutation [30]. Mechanism sparsity regularization offers another route; by enforcing sparsity on the learned adjacency matrices for auxiliary influences and temporal dependencies, methods can identify representations up to consistency equivalence, effectively disentangling factors even in dense graphs [31, 32].

Concept-free approaches like CCVGAE (Concept-free Causal VGAE) prove a tight upper bound for approximating optimal latent factors via linear causal modeling without requiring concept labels, demonstrating that linear causal structures can emerge from unsupervised graph data [33]. Furthermore, C-Disentanglement (cdVAE) introduces a `do-c` operator to estimate interventional distributions using observable confounder proxies, partitioning data to enforce local statistical independence that implies global causal disentanglement [34].

2.2.3 Temporal and Hierarchical Identifiability Temporal data provides natural constraints for identification. TDRL (Temporally Disentangled Representation Learning) factorizes unknown distribution shifts into transition and observation change factors, proving identifiability for nonparametric latent causal processes under linear independence conditions of partial derivatives [35]. For hierarchical structures, CHiLD (Causally Hierarchical Latent Dynamic) proves that $2L+1$ temporal observations are required for block-wise identifiability of L-layer hierarchies, utilizing normalizing flow priors to model complex dynamics [36].

In the context of instantaneous dependencies, IDOL (IDentification framework for instantaneOus Latent dynamics) imposes a sparse influence constraint on time-delayed and instantaneous relations, enabling identifiability up to a Markov equivalence class using only temporal contextual information [37]. Additionally, TRACE models causal mechanisms as convex combinations of atomic bases, recovering continuous mixing trajectories via geometric projection, which aids in identifying mechanisms that evolve continuously over time [38].

2.3 Causal Generative Modeling and Counterfactual Inference

Generative models are essential for simulating interventions and generating counterfactuals, which correspond to the third rung of the causal ladder. Deep Structural Causal Models (DSCMs) integrate deep learning mechanisms into SCMs to enable tractable counterfactual inference [39].

2.3.1 Normalizing Flows and Invertible Mechanisms Normalizing flows are particularly suited for causal modeling due to their invertibility, which is required for the "abduction" step in counterfactual reasoning (inferring exogenous noise from observations). Deep Causal Graphs (DCG) specify an abstract contract for neural networks to model causal distributions, utilizing Conditional Normalizing Flows for flexible conditional probability densities [40]. Causally Consistent Normalizing Flow (CCNF) translates SCM DAGs into ordered sequences via topological batching, constructing chains of partial causal transformations that maintain consistency with the causal graph [41].

For scenarios with only causal ordering known, P-CFM (Parallel-Causal Flow Model) constructs Triangular Monotonically Increasing maps via Probability Flow ODEs, proving identifiability without full graph topology and enabling parallel abduction [42]. FlexCausal employs block-diagonal VAEs with independent Normalizing Flows for each causal concept, decoupling mechanism learning from distributional statistics to model complex non-Gaussian exogenous noise [43].

2.3.2 Diffusion Models for Causal Generation Diffusion models have recently been adapted for causal tasks. CausalDiffAE integrates structural causal models into diffusion autoencoders, parameterizing causal mechanisms among latents to enable isolated interventions on semantic factors while preserving downstream consistency [44]. DiffusionCounterfactuals guides the reverse diffusion process with gradients derived from a Neural Structural Causal Model (NSCM), ensuring generated samples adhere to intervened factors and maintaining causal consistency [45]. DCRL (Diffusion-based Causal Representation Learning) utilizes diffusion-based infinite-dimensional latent codes to encode causal information equivalent to noise variables in SCMs, showing superior performance in higher dimensions compared to VAE-based baselines [46].

2.3.3 Counterfactual Generation and Fairness Generating valid counterfactuals is critical for fairness and explainability. DeepBC (Deep Backtracking Counterfactuals) formulates counterfactual generation as a constrained latent optimization problem, minimizing the distance to exogenous variables while satisfying antecedent constraints, thus ensuring causal compliance [47]. VCI (Variational Causal Inference) derives an individual-level Evidence Lower Bound (ELBO) for counterfactual likelihood, optimizing factual reconstruction and latent divergence to handle high-dimensional outcomes [48].

In fairness applications, DCEVAE (Disentangled Causal Effect Variational Autoencoder) partitions exogenous uncertainty into intervention-independent and intervention-correlated components, enabling precise counterfactual fairness by disentangling the effects of sensitive attributes [49]. Causal Manifold Fairness (CMF) enforces fairness by minimizing the difference between the Pullback Metric Tensor and the Hessian of factual counterfactuals, formalizing fairness as geometric invariance on the data manifold [50].

2.4 Temporal and Dynamic Causal Discovery

Time-series data introduces unique challenges and opportunities, such as lagged effects, non-stationarity, and Granger causality.

2.4.1 Neural Granger Causality and Time-Series Models Neural Granger Causality methods replace linear autoregressive models with deep networks. LeKVAR (Learned Kernel VAR) parameterizes kernel functions via shared neural networks, eliminating manual kernel selection and improving estimation precision [51]. GRNGC (Gradient Regularization-based Neural Granger Causality) infers causality via input-output gradients of a unified forecasting model, providing a more robust signal than static weight magnitudes [52]. Similarly, JRNGC utilizes an input-output Jacobian regularizer to enforce sparsity on variable dependencies in a single multivariate forecasting model [53].

For irregular time series, CUTS and CUTS+ employ iterative alternating optimization between data imputation and causal graph fitting, utilizing Delayed Supervision GNNs and coarse-to-fine discovery strategies to handle missing data and high dimensionality [54, 55]. CausalFormer integrates a causality-aware transformer with Regression Relevance Propagation to decompose model contributions, achieving state-of-the-art performance in temporal causal discovery [56].

2.4.2 Non-Stationarity and Dynamic Structures Non-stationary environments, where causal mechanisms change over time, require adaptive models. LiLY (Learning Latent causal dYnamics) partitions latent space into fixed, changing, and observational subspaces, exploiting distribution changes as soft interventions to identify time-delayed latent variables [57]. NCTRL (Nonstationary Causal Temporal Representation Learning) models non-stationary states as a discrete Markov process, enabling identifiability of latent components without auxiliary domain labels [58].

State-Dependent Causal Inference (SDCI) models conditionally stationary dynamics via discrete latent state variables, introducing conditional summary graphs to compactly represent state-dependent edges [59]. For continuous mechanism evolution, TRACE models mechanisms as convex combinations of atomic bases, recovering trajectories via geometric projection [38]. Additionally, LOCAL (Learning with Orientation matrix) formulates a quasi-maximum likelihood score for dynamic causal structure learning, decoupling causal masks from real matrices to improve explainability [60].

2.4.3 Event Sequences and Point Processes For event-based data, TNPAR (Topological Neural Poisson Auto-Regressive) integrates prior topological networks into neural Poisson processes, correcting spurious edges caused by non-i.i.d. dependencies in event sequences [61]. MCNS (Mining Causal Natural Structures) discovers causal relationships inside univariate time series subsequences via snippet clustering and GFCI, injecting causal strength into prediction models [62].

2.5 Integration of Prior Knowledge and Foundation Models

Incorporating domain knowledge and leveraging foundation models are emerging trends to improve the reliability and scalability of causal discovery.

2.5.1 Knowledge-Guided and Hybrid Approaches Injecting expert knowledge can constrain the search space and improve accuracy. CASTLE regularizes neural networks via auxiliary causal graph discovery, reconstructing only causally connected features to improve generalization [63]. CINN (Causality-Informed Neural Network) maps causal DAG nodes directly to neural units, enforcing strict causal orientation via connectivity constraints [64]. Knowledge-constrained Bayesian network learning integrates hard and soft constraints (e.g., forbidden edges, temporal orders) into structure learning algorithms, significantly reducing false positives [65].

Hybrid approaches combine data-driven learning with symbolic reasoning. HCP-DCNet decomposes causal scenes into typed primitives, fusing symbolic logic constraints with sub-symbolic attention via a Causal Flow Conservation Principle [66]. Causal-Symbolic Meta-Learning (CSML) meta-learns a shared causal graph structure as a structural prior, enabling rapid adaptation to novel tasks [67].

2.5.2 Large Language Models for Causal Reasoning LLMs are increasingly used to extract causal priors or guide discovery. GPT-4 Guided Causal ML extracts causal edges from variable labels using LLMs and converts them into structural constraints for score-based learners, demonstrating performance indistinguishable from domain experts in small networks [68]. ILS-CSL (Iterative LLM Supervised CSL) employs an iterative loop where an LLM verifies edge directionality, reducing error propagation compared to full pairwise inference [69].

LLMs are also used for representation elicitation. COAT (Causal representatiOn AssistanT) uses an iterative propose-verify loop where LLMs propose high-level factors from unstructured data, which are then analyzed by causal discovery algorithms to construct feedback [70]. Furthermore, TabPFN embeddings have been shown to encode causal structures beyond statistical correlations due to pre-training on synthetic SCMs, allowing for the extraction of implicit causal knowledge via adapter frameworks [71].

2.5.3 Causal Representation in Specific Domains In biomedicine, SENA-discrepancy-VAE embeds Gene Ontology processes as structural priors to recover biologically meaningful causal factors from single-cell data [72]. LLM4GRN integrates LLMs with statistical causal inference to infer gene regulatory networks, utilizing LLM internal knowledge for transcription factor identification [73]. In climate science, DAG-VAE embeds physics-informed DAGs within VAE latent spaces to disentangle overlapping teleconnection signals like ENSO and IOD [74]. In robotics, CausalCF adapts counterfactual architectures for RL environments, enabling learning of complete SCMs from scratch to improve robustness [75].

3. Challenges and Outlook

Despite significant progress, several critical challenges remain in causal representation learning and discovery.

3.1 Identifiability and Assumption Dependence

A recurring limitation across many methods is the reliance on strong, often untestable assumptions. Identifiability proofs frequently require linearity, additive noise, non-Gaussianity, or specific intervention types (e.g., perfect single-node interventions) that may not hold in real-world scenarios [3, 6]. For instance, many latent causal models assume the absence of unobserved confounders or specific graph structures (e.g., bow-free) which are difficult to verify empirically [76, 21]. The "Predictive-Causal Gap" theorem highlights that population risk minimizers in predictive learning often tilt towards predictable environment modes rather than causal system modes, suggesting that standard self-supervised objectives may inherently fail to learn causal representations without explicit causal grounding [77]. Future research must focus on developing methods with weaker assumptions or robustness to assumption violations.

3.2 Scalability and Computational Complexity

While continuous optimization has improved scalability, many methods still struggle with high-dimensional data. The computational cost of evaluating acyclicity constraints, computing Jacobians, or performing MCMC sampling scales poorly with the number of variables [78, 5]. Methods like Enes and CauScale address this via amortization and dimensionality reduction, but general scalability to genome-scale or pixel-level causal discovery remains an open problem [15, 79]. Additionally, the integration of LLMs introduces significant computational overhead and latency, particularly in iterative refinement loops [68, 70].

3.3 Evaluation and Benchmarking

The lack of standardized benchmarks and ground truth in real-world data hinders fair comparison and rigorous evaluation. Most evaluations rely on synthetic data or small-scale biological datasets (e.g., Sachs, Perturb-seq) where the ground truth is assumed known [5, 80]. Real-world causal effects are rarely observable, making it difficult to validate counterfactual predictions [81]. Recent efforts like CausalTime and CausalDynamics aim to generate more realistic synthetic benchmarks, but there is a pressing need for large-scale, real-world datasets with verified causal structures [82, 83].

3.4 Integration with Foundation Models and Unstructured Data

While LLMs show promise in extracting causal knowledge, their reliability and hallucination risks pose challenges. The alignment between LLM-generated causal graphs and statistical evidence from data is not always guaranteed, and methods for effectively fusing these disparate sources of information are still nascent [68, 73]. Furthermore, applying causal discovery to unstructured data (text, images) requires robust representation learning pipelines that can bridge the gap between low-level features and high-level causal concepts [84, 85].

3.5 Safety, Ethics, and Deployment

Deploying causal models in high-stakes domains like healthcare and finance raises safety and ethical concerns. Errors in causal structure learning can lead to biased policy decisions or harmful interventions [86]. Ensuring that causal models are robust to adversarial attacks and distribution shifts is crucial for safe deployment [87]. Additionally, the interpretability of deep causal models, while better than black-box predictors, still requires further development to be actionable for domain experts [88].

4. Conclusion

Causal Representation Learning and Causal Discovery in deep learning represent a frontier in artificial intelligence, promising models that are robust, interpretable, and capable of reasoning about interventions and counterfactuals. This survey has synthesized a vast body of literature, highlighting the transition from discrete search to continuous optimization, the theoretical advances in identifiability via interventions and temporal dynamics, and the emergence of causal generative models. We have seen how methods like differentiable DAG learning, invariant representation learning, and causal normalizing flows are reshaping the landscape.

However, the path to truly causal AI is fraught with challenges. The dependence on strong assumptions, the scarcity of real-world benchmarks, and the computational demands of exact inference remain significant hurdles. Future research directions should prioritize the development of assumption-robust algorithms, the creation of comprehensive real-world causal benchmarks, and the seamless integration of causal reasoning with foundation models. By addressing these challenges, the community can move towards AI systems that not only predict the future but understand the mechanisms that drive it, enabling safer and more reliable decision-making in complex, dynamic environments.

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